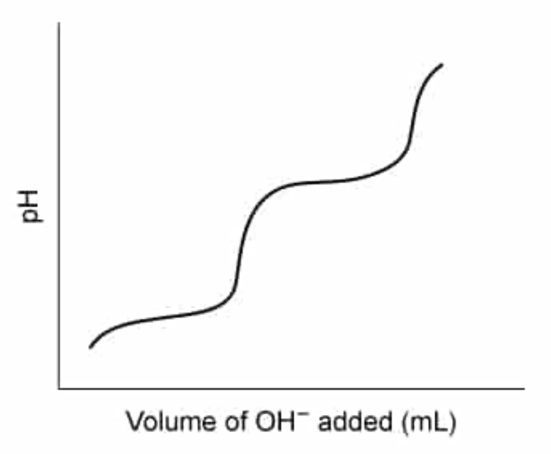
The figure below shows the resulting titration curve for an acid titrated with aqueous sodium hydroxide.

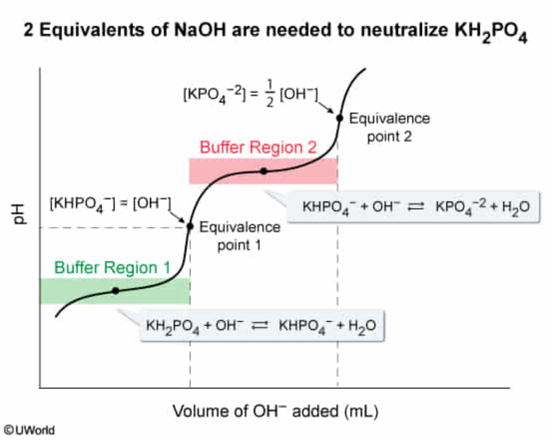


When titrated in solution, which of the following salts would be most likely to produce a similar titration curve?

- A. H_3PO_4 [15%]
- B. KH_2PO_4 [67%]
- C. Na_2HPO_4 [11%]
- D. K_3PO_4 [4%]

Correct
68% answered correctly

Explanation:



In the titration of an acid, a measured amount of a **base** solution (titrant) with a known concentration is slowly added to another solution that contains an unknown concentration of an acid. As the titration proceeds, acidic protons (weakly bound hydrogen atoms that easily ionize into H^+ ions) react with the base in an acid-base neutralization reaction, which slowly changes the pH of the acid solution being titrated.

During neutralization, if an acid structure has more than one acidic proton, the acidic protons dissociate one at a time in a stepwise progression from the most acidic to the least acidic proton. Each step in the progression generates a new conjugate base, which functions together with the remaining acid as a buffer that resists changes in pH (seen in a titration curve as relatively flat regions).

An equivalence point is reached when the H^+ ions from a given intermediate acidic species are fully neutralized. On a titration curve, this is signaled by a rapid change in pH, and the equivalence point is located at the middle of this nearly vertical segment. Accordingly, the titration curve of an acid shows one **buffer region** and one equivalence point **for each acidic proton** in its structure. When enough OH^- titrant is added to consume all the acidic protons (ie, the solution is fully neutralized), adding more OH^- will have no effect on the titration curve and the curve will flatten as the pH of the solution reaches the pH of the titrant.

In this question, the given titration curve exhibits two buffer regions and two equivalence points. As such, the titrated acid must have two acidic protons. Of the four compounds listed, only KH_2PO_4 is diprotic:

$$KH_2PO_4 \rightleftarrows 2\,H^+ + K^+ + PO_4^{3-}$$

Therefore, KH_2PO_4 would produce a titration curve similar to that given in the question.

(Choice A) H_3PO_4 would exhibit three equivalence points in a titration because it has three acidic protons:

$$H_3PO_4 \rightleftarrows H^+ + H_2PO_4^-$$

$$H_2PO_4^- \rightleftarrows H^+ + HPO_4^{2-}$$

$$HPO_4^{2-} \rightleftarrows H^+ + PO_4^{3-}$$

(Choice C) Na_2HPO_4 would exhibit just one equivalence point in a titration because it has only one acidic proton:

$$Na_2HPO_4 \rightleftarrows H^+ + 2\,Na^+ + PO_4^{3-}$$

(Choice D) K_3PO_4 is not acidic (has no acidic protons) and would show no equivalence points in a titration with a base.

Educational objective:
During the titration of an acid with a sufficiently strong base, the titration curve exhibits one buffer region and one equivalence point for each acidic proton in the acidic compound structure.

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